

Exploring BMI and Physical Activity in Students

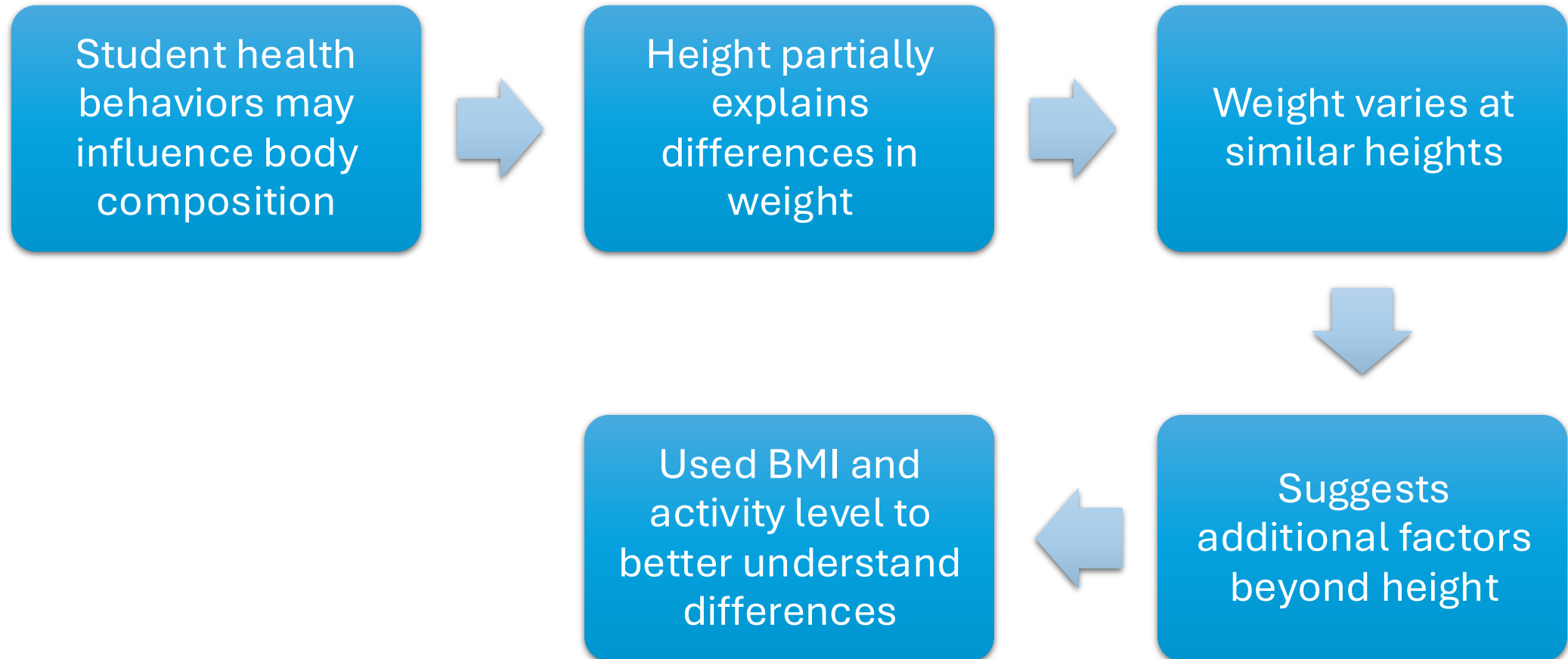
By: Arianna Childs and
Halden Kerzner

Does physical health explain differences in BMI beyond what height alone can explain?

Data Sources and Variables

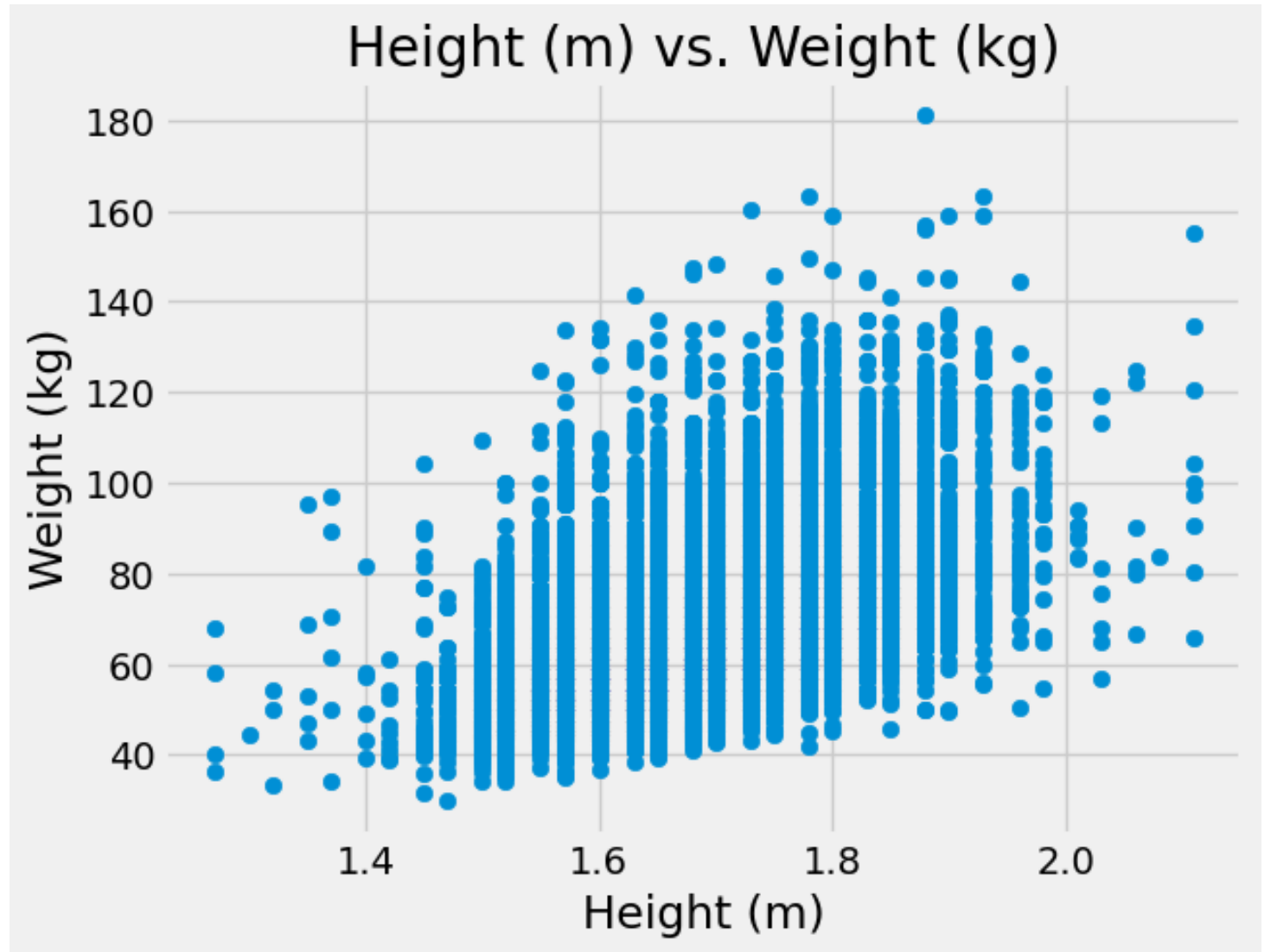
- Youth Risk Behavior Surveillance System (YRBSS): high school behavior survey
 - Height(m), weight(kg), physical activity(days/week)
- Kaggle Dataset
 - BMI and activity hours per week
- Body Mass Index (BMI) used as combined measure to estimate a person's body fat:
 - $BMI = \text{weight(kg)} / (\text{height} * \text{height})(m)$

Motivation for Research



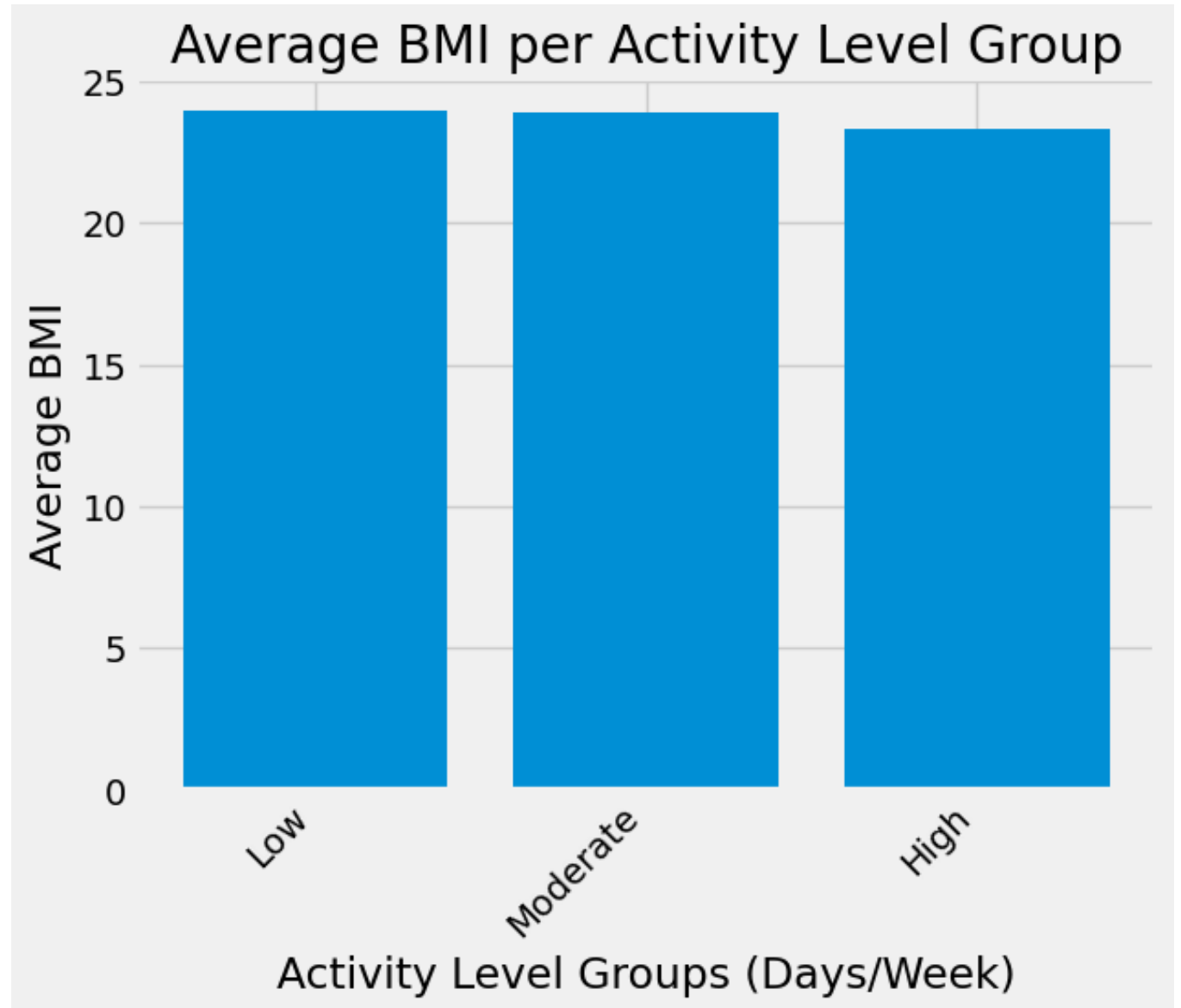
Students of Similar Height Often Show Different Weights

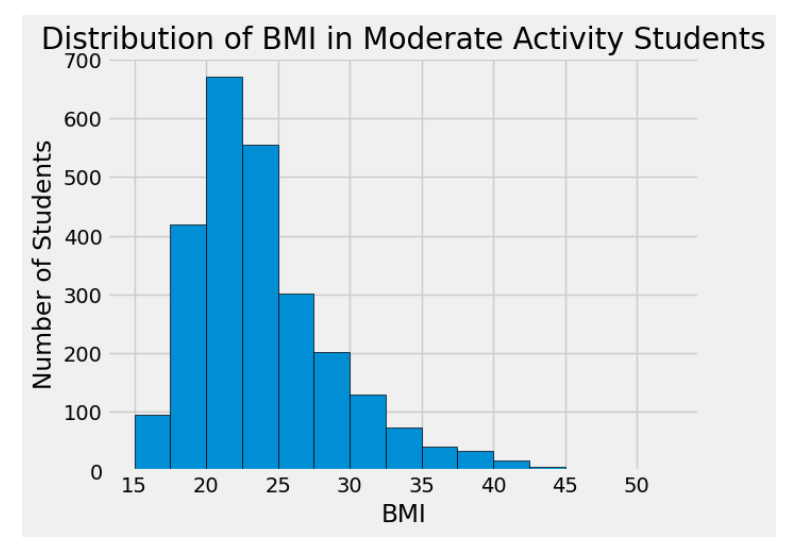
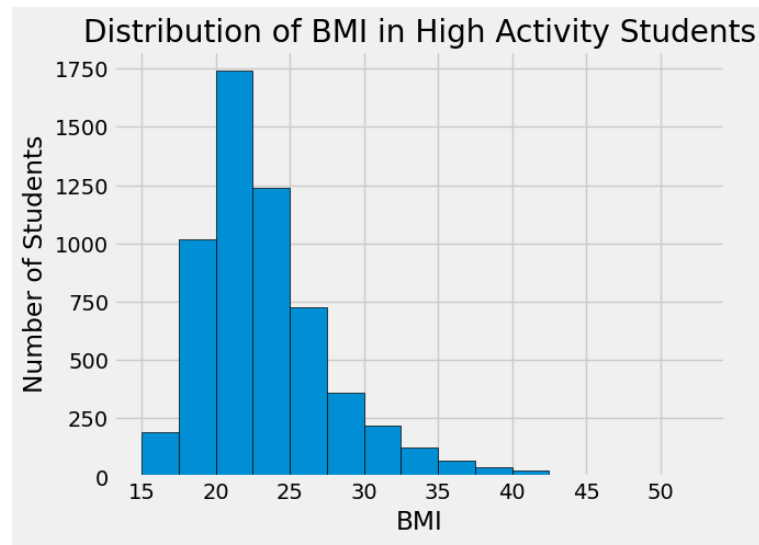
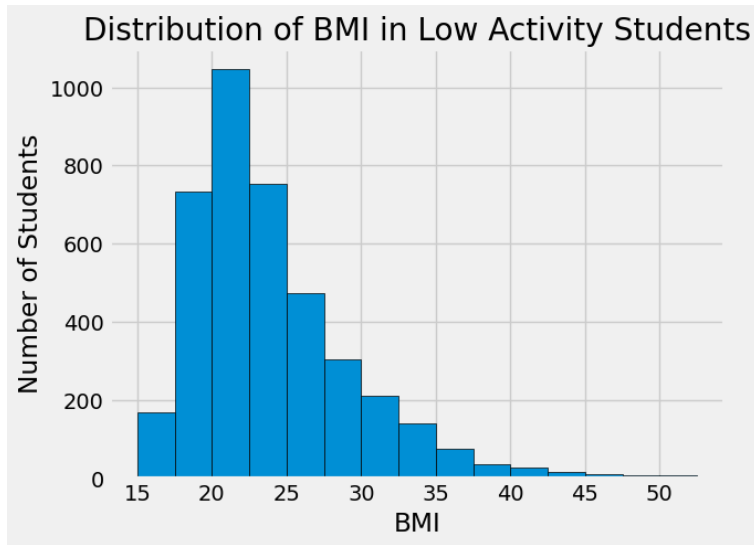
- Moderate positive relationship between height and weight
- Correlation Coefficient $r \approx 0.51$
- Students of similar heights have noticeable variations in weight
- Height alone does not fully explain differences in weight
- Led to using BMI to better understand weight differences, examining how activity level relates.



Higher Activity Groups Show Slightly Lower Average BMI

- Average BMI is similar across groups
- Slight decrease in BMI with high activity group
- Differences are small
- Led to examining distributions



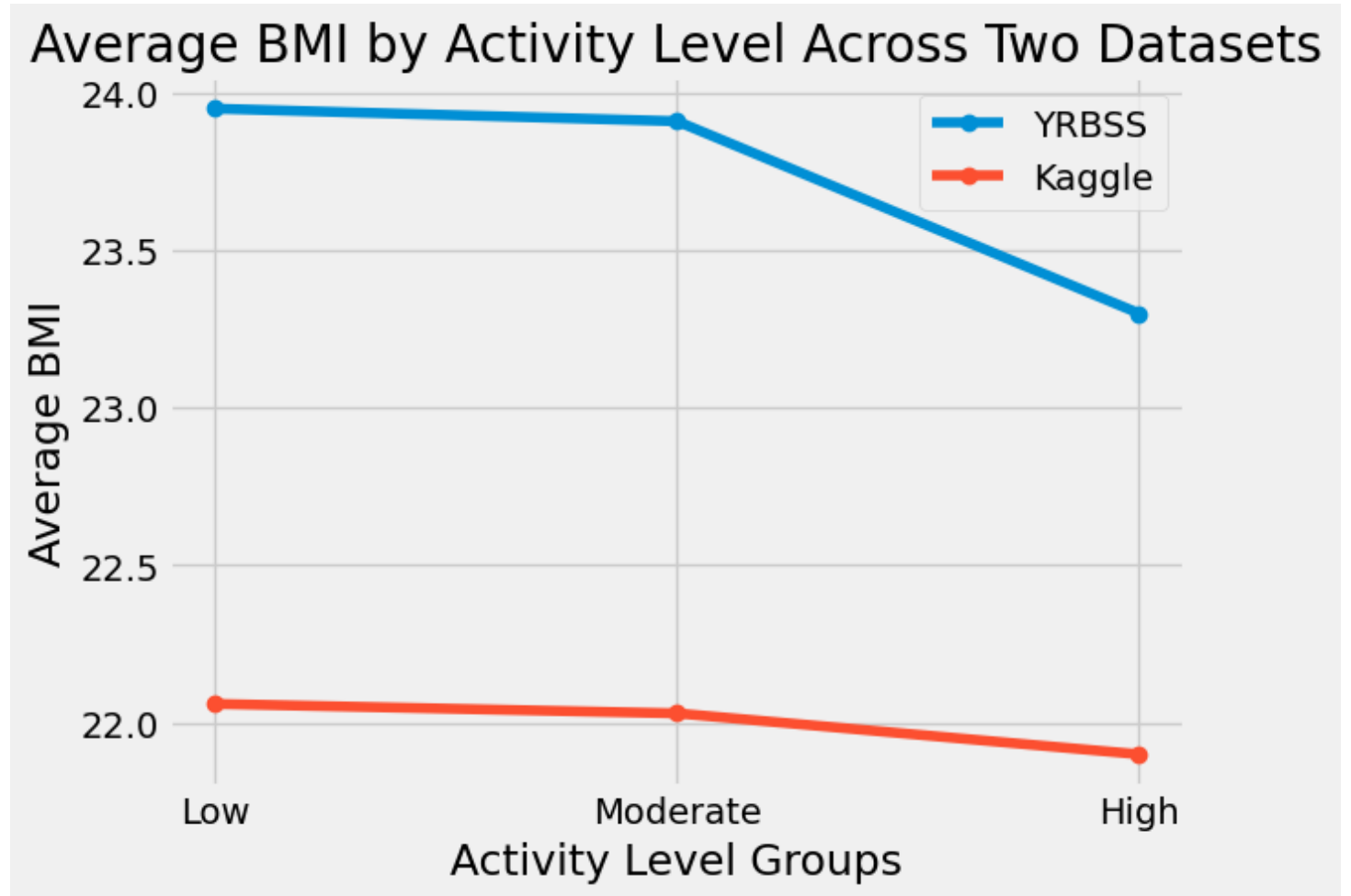


Distributions Reveal Patterns Hidden by Averages

- Strong concentration of students around BMI 20-25
- Low activity has more high BMI extremes
- High activity shifts slightly toward lower BMI
- Histograms reveal extremes not considered in averages

Are These Patterns Consistent Across Other Datasets?

- Both datasets show lower BMI in high activity group
- YRBSS decline strong
- Kaggle trend smaller but consistent
- Suggests findings may generalize



Key Findings

- Height alone does not explain weight variation
- BMI provides stronger health context
- Higher activity groups tend to show slightly lower BMI
- Similar pattern appears in two independent datasets



Future Work

- Include diet and nutrition variables
- Include activity intensity and duration, not just frequency
- Account for lean mass in evaluating BMI
- Incorporate body composition measures beyond BMI
- Study BMI patterns by gender/age subgroups
- Examine whether similar patterns exist in other age groups

Lessons and Challenges

Lessons:

- Data cleaning
- Subgroup definitions affect results
- Dataset joins can strengthen conclusions

Challenges:

- Two different activity measures (days/week vs hours/week)
 - Handling missing data / grouping continuous variables
 - Structuring a clear research process from initial observations
 - Translating raw data into meaningful patterns
 - Interpreting subtle differences in results
- 